

OUTPUT 1:

```
=====
          ARITHMETIC OPERATIONS PROGRAM
=====
Enter first integer: 15
Enter second integer: 30

--- ADDITION & SUBTRACTION ---

--- ADDITION ---
Adding 15 + 30
Result = 45

--- SUBTRACTION ---
Subtracting 15 - 30
Result = -15

--- MULTIPLICATIONS (Final Answer) ---

Sequential Multiplication: 450
Bit Multiplication: 450
Bit-Pair Multiplication: 450
Booth's Multiplication: 450

--- DIVISIONS (Final Answer) ---
Restoring Division: Quotient=0, Remainder=15
Non-Restoring Division: Quotient=0, Remainder=15
```

OUTPUT 2:

```
=====
          ARITHMETIC OPERATIONS PROGRAM
=====
Enter first integer: 85
Enter second integer: -20

--- ADDITION & SUBTRACTION ---

--- ADDITION ---
Adding 85 + -20
Result = 65

--- SUBTRACTION ---
Subtracting 85 - -20
Result = 105

--- MULTIPLICATIONS (Final Answer) ---

Sequential Multiplication: -1700
Bit Multiplication: -1700
Bit-Pair Multiplication: -1700
Booth's Multiplication: -1700

--- DIVISIONS (Final Answer) ---
Restoring Division: Quotient=-5, Remainder=-15
Non-Restoring Division: Quotient=-5, Remainder=-15
```